

Training: gru

A 16-unit GRU over the RR sequence plus the 9 features.

The GRU is the slowest network to train per step: its 30 time steps are unrolled, so each step runs 30 recurrent updates forward and backward.

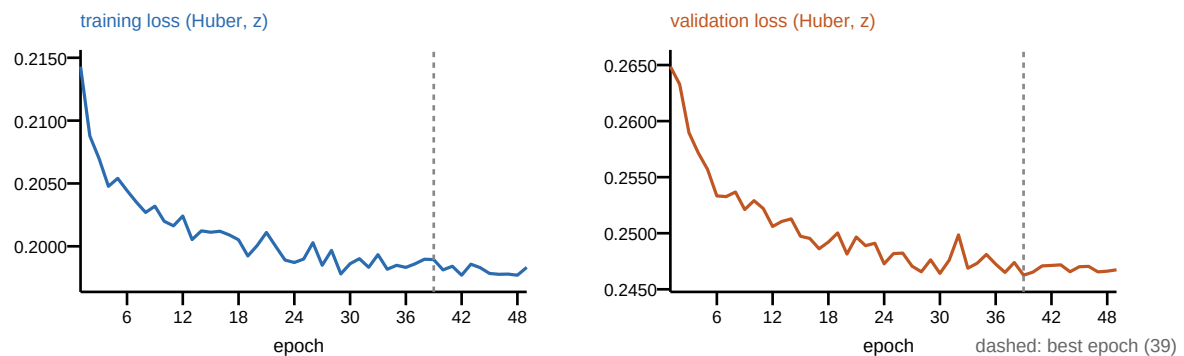
How this family is trained

- GRU(16, unroll=True) over the (30, 2) sequence, its last state concatenated with the 9 'seq' features, then Dense(32, ReLU) - Dense(1); residual target; the shared Keras recipe with at most 90 epochs.
- Before the run, a throughput probe compared 64, 128 and 256 steps per Keras call: 18-24, 24 and 23 ms per step, so 64 was kept (the cost is on the device, not in the dispatch).

Runs

gru - round 3

item	value
command	python src/train.py --models gru --epochs 90 --tag gru
start	from scratch (random initialisation, seed 211)
inputs	30-beat RR sequence (z-RR, z-dRR) + 9 tabular features (set seq)
target	z-scored correction over the RLS forecast (dRR - rls_pred)
epoch	1,999,872 windows drawn at random with replacement (512 x 3906 steps)
budget	49 of at most 90 epochs (early stopping, patience 10); 97,993,728 windows drawn = 4.7 passes over the training set
learning rate	0.001 at the start, halved 6 times on validation plateaus, 1.6e-05 at the end
best epoch	39 (validation loss 0.2649 after epoch 1, 0.2463 at best); its weights are the saved model
time	1.32 h on one GPU, 24.7 ms per step
int8	min/max calibration on 20,000 random training windows; first 20k validation windows: 35.05 vs 34.02 ms float; full validation: int8 34.81 vs float 33.90 ms (+2.71%)
validation RMSE (float)	33.90 ms on all 1,421,225 validation windows (-8.7% vs rls_ar 37.14)



- A first round-1 attempt was interrupted; its folder runs/gru_interrupted was kept but holds no model or config. This run was queued on 2026-10-07 at 21:46 and is the only gru model.
- Exported with the 20,000-window calibration (trained after the calibration fix, before best-of-K).

What the training showed

- In float the GRU is competitive (it ties micro_tcn), but int8 quantisation of the recurrent state costs it +2.71% on validation, over the 2% limit, and 1.6-3.6 ms in the closed loop: rejected. Better training would not fix the quantisation.

Abbreviations and acronyms used in this document

term	meaning
dRR	beat-to-beat change of the RR interval: $dRR[n+1] = RR[n+1] - RR[n]$ (ms); the quantity predicted
GRU	gated recurrent unit (a recurrent neural-network cell)
gru	gated recurrent unit (run name)
HRV	heart rate variability
int8	8-bit integer: weights and activations quantised to 8-bit integers, as run on the microcontroller; _int8 = the int8 version of a run
ms	milliseconds
ReLU	rectified linear unit, $\max(0, x)$ (an activation function)
RR	RR interval: time between two consecutive heartbeats (R waves of the ECG), in ms
seq	feature-set / model-kind name: sequence model (takes the raw RR window plus tabular features)
tcn	temporal convolutional network (run-name part)